



Investigating the impact of supply chain management practices on delivery dependability

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ABSTRACT

This paper explores the impact of Supply Chain Management (SCM) practices on delivery dependability. Our research considers five dimensions of SCM practices, namely strategic supplier partnership, customer relationship, information sharing, information quality and internal lean practices. By reviewing strategic performance theory, we examine the impact of SCM practices on delivery dependability, the extent to which an organization is capable of providing on time, the ordered type and volume of product in conformance with customer specifications. We also examine the role of SCM information systems on the application of SCM practices across supply chain partners. The empirical findings from a survey of 47 manufacturing firms in Greece confirmed the crucial role of all SCM practices and especially information sharing on the establishment of a sustainable competitive advantage boosting supply chain performance outcomes. Managerial implications are discussed.

Categories and Subject Descriptors

J.1 [Computer Applications]: Administrative Data Processing – Business

General Terms

Management

Keywords

Supply Chain Management, SCM practices, SCM information systems, information sharing, delivery dependability

1. INTRODUCTION

A supply chain is a network that consists of suppliers, manufacturers, warehouses, distributors and retailers who coordinate their plans and activities in order to convert raw materials to finished goods. Last decades, organizations have realized that in order to face the challenge of global economy and increased customer demands they have to make competitive their entire supply chain. Supply Chain Management (SCM) is defined as “the systemic, strategic coordination of the traditional business

functions and tactics across these businesses functions within a particular organization and across businesses within the supply chain for the purposes of improving the long-term performance of the individual organizations and the supply chain as a whole” [1].

The main contribution of SCM to organizations is the offering of the means to link technology with people in an attempt to align the technology with the capabilities of each organization and among its trading partners [6][4]. The most important supply chain processes include product development, procurement, manufacturing, physical distribution, customer relationship management and performance measurement [5].

In this paper, we explore the impact of SCM practices on delivery dependability, an attribute that influences customer satisfaction. We present the methodology and findings of a field research that was conducted in 2013, in 47 manufacturing firms at the region of Central Greece. The aim of this survey is the investigation of the impact of five dimensions of SCM practices on delivery dependability. These dimensions include strategic supplier partnership, customer relationship, information sharing, information quality, and internal lean practices [3].

The rest of the paper is organized as follows: Section 2 presents the methodology of the field research that was conducted, the data analysis and the results that were revealed. Section 3 discusses the research results and, section 4 concludes the paper.

2. METHODOLOGY

2.1 Questionnaire Design and Sampling

The field research was based on a structured questionnaire. It was built by adapting existing scales in the SCM literature measuring relevant practices, techniques and methods, as well as Supply Chain delivery dependability.

For the purpose of comprehensively capturing the aspects of SCM practices, our construct was built on the basis of several criteria, which are conceptualized and used in previous empirical studies [2]. SCM practices' scale consists of 8 items, specified in a 7-point Likert scale and has been validated in similar contexts to those in our study.

Principal Component Analysis (PCA) with normalized varimax rotation was performed and five factors were extracted from the SCM practices' measure, explaining approximately 72% of the overall variance. Preceding PCA, the Bartlett sphericity testing on the degree of correlation between the variables ($p < 0.001$) and the Kaiser–Meyer–Olkin (KMO) index verified the appropriateness of the sample. Cronbach's coefficient alpha was calculated to test internal reliability for each principal component ranging from 0.77 to 0.91. Thus, reliability coefficients for all dimensions

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proved to be well over the minimum acceptable reliability level of 0.7.

The field research was conducted in 2013 to test the impact of several SCM practices on delivery dependability, in a cross-sectional sample of firms at the region of Central Greece. The final sample consisted of 200 manufacturing SMEs with 10 to 250 employees.

The questionnaire was tested twice before it was released, by ten logistics managers and by academics for in depth discussions. They confirmed the cognitive relevance of the questionnaire to the Supply Chain of manufacturing firms. The questionnaires were distributed to logistics managers or top executives. From the initial sample of 200 enterprises, 47 firms responded achieving a response rate of 23.5 %, which is considered to be satisfactory for this type of empirical research.

2.2 Data Analysis and findings

The mean scores, standard deviations and t-test of the differences between the mean and mid-point (at a 7-point Likert scale, mid-point=4) were estimated. The standard deviations for each of the scores indicate that respondents' views varied considerably across the sample. Overall, the difference between the mean scores and the mid-point indicates that respondents perceived that they extensively use practices facilitating strategic supplier partnership (mean diff.= 1.254, $p<0.001$), customer relationship (mean diff.= 1.559, $p<0.001$), information sharing (mean diff.= 1.130, $p<0.001$), information quality (mean diff.= 1.342, $p<0.001$), and internal lean practices (mean diff.= .847, $p<0.001$). Also, participants exhibit high levels of performance referring to delivery dependability (mean diff.= 1.444, $p<0.001$).

Moreover, correlation analysis was conducted. The bi-variate relationships (Pearson) between the various SCM practices and delivery dependability of the firms participating were estimated. The results show that all groups of practices (SSP: $r=0.287$, $p<0.05$, CR: $r=0.425$, $p<0.001$, IS: $r=0.410$, $p<0.001$, IQ: $r=0.488$, $p<0.001$, and ILP: $r=0.444$, $p<0.001$) have significant influence on firm's delivery dependability performance.

3. DISCUSSION

This study investigates the relationship between SCM practices on organizational performance measured by delivery dependability. Drawing from a sample of micro and small manufacturing enterprises located at Central Greece, all types of SCM practices namely, strategic supplier partnership, customer relationship, information sharing, information quality and internal lean practices, proved to exert a significant effect on delivery outcomes.

Our results illuminate the important role of the effective implementation of practices fostering supply chain partners' coordination and cooperation and subsequently leading to increased organizational performance. Information quality practices reflecting the accuracy, timeliness, adequacy, and credibility of information exchanged. Information sharing methods provide the necessary foundation for the backward coordination of information technology and the forward coordination of the physical flow of deliveries.

Similarly, given that collaboration's starting point lies at the customers' side and it follows the chain back through the firm to suppliers, internal integration guided by internal lean practices as well as external integration are crucial success factors. On the other hand, external integration of the supply chain partners can be achieved by investing on nurturing customer relationships and strategic partnerships with suppliers.

To summarize, SCM practices targeting strategic supplier partnership, customer relationship, information sharing and quality, and internal lean operations provide the necessary ground to SCI, which leads to increased organizational performance.

4. CONCLUSIONS

Nowadays, Supply Chain Management (SCM) has to deal with increased customer demands and global competition at the same time. The evolution of SCM practices and techniques is a factor that enabled the integration of supply chains into value systems.

In order to assess the competitive value of a variety of practices and methods for SCM, we have conducted a field research in 2013, in a cross-sectional sample of firms at the region of Central Greece. The field research used a structured questionnaire, which was built on the basis of several criteria used in previous empirical studies. The results confirm the crucial role of SCM practices and techniques on the establishment of a sustainable competitive advantage based on delivery dependability.

For the future, our conceptual framework may also be replicated in other regions of Greece, in order to compare our findings with these of the current research. In addition, the impact of contextual factors such as business environment, on the link between SCM practices and competitive advantage should be explored.

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